

A Minimal State–Structure Model for Dynamical Structural Selection

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Abstract

We study a minimal phenomenological state–structure model based on a quartic Landau free energy with three scalar order parameters. The model is analysed analytically and numerically by identifying all stationary points and selecting the global minimum.

A systematic parameter scan in (λ, η) is performed using multi-start root finding combined with symmetry reduction. The resulting phase structure is classified into aligned and frustrated sectors depending on the transverse order parameter v .

We find that, within the present quartic model, the phase boundary separating frustrated and aligned regimes satisfies $\eta_c(\lambda) = 0$. This follows from the separable structure of the reduced free energy and is reproduced by the numerical scan.

The model serves as a minimal reference system for testing numerical branch-tracking and phase-selection procedures in more complex state–structure theories. A second-level formulation based on a full qubit state–structure functional is outlined as the basis for future quantitative work.

1 Conceptual Background

We start from the formal structural decomposition

$$L_{\text{total}} = L_{\text{physics}} + L_{\text{structural}} + L_{\text{coupling}}, \quad (1)$$

where structural contributions encode coherence, balance, and alignment. The effective structural free energy takes the schematic form

$$F_{\text{struct}} = E_{\text{phys}} + \alpha \cdot S_{\text{entropy}}, \quad (2)$$

and a generic structural score is written as

$$\hat{C} = \frac{H \cdot B \cdot S \cdot V \cdot R}{\Delta M + 0.01}. \quad (3)$$

The associated structural selection principle is

$$P(\phi_i) \propto \exp(-F_{\text{struct}}/T), \quad \text{CCI} = \frac{\text{destabilizing}}{\text{stabilizing}}. \quad (4)$$

The paper is organised in two levels. Section 2 presents a quartic reduction that is analytically transparent and directly implemented in the companion numerical code. Section 3 presents the physically richer qubit state–structure theory, which motivates the Level-0 reduction and provides the basis for future quantitative analysis.

2 Level 0: Quartic Phenomenological Toy Model

2.1 Model Definition

To analyse the qualitative phase structure in a minimal setting, we consider

$$F_{\text{red}}(\lambda, \eta; m, u, v) = -\lambda(m^2 + u^2) + \eta v^2 + m^4 + u^4 + v^4, \quad (5)$$

with control parameters λ and η . The longitudinal amplitudes (m, u) couple to λ ; the transverse amplitude v couples to η and measures structural frustration.

2.2 Analytic Structure

The stationarity conditions $\nabla F_{\text{red}} = 0$ read

$$-2\lambda m + 4m^3 = 0, \quad (6)$$

$$-2\lambda u + 4u^3 = 0, \quad (7)$$

$$2\eta v + 4v^3 = 0. \quad (8)$$

These equations decouple: λ controls (m, u) and η controls v independently. The exact solutions are

$$m, u \in \{0, \pm\sqrt{\lambda/2}\} \quad (\lambda > 0), \quad v \in \begin{cases} \{0\} & \eta \geq 0, \\ \{0, \pm\sqrt{-\eta/2}\} & \eta < 0. \end{cases} \quad (9)$$

Thus the exact phase boundary is

$$\eta_c(\lambda) = 0, \quad (10)$$

independent of λ . The quartic reduction is therefore exactly solvable, and its phase structure is governed entirely by $\text{sign}(\eta)$.

2.3 Phase Classification

- *Aligned sector* ($\eta \geq 0$): only $v = 0$ is stable; the transverse degree of freedom is energetically suppressed.
- *Frustrated sector* ($\eta < 0$): nonzero minima $|v| = \sqrt{-\eta/2} > 0$ appear, while the stationary point $v = 0$ ceases to be the stable minimum in the transverse sector.

The parameter λ controls the magnitude of the longitudinal ordered phase but does not shift the phase boundary.

2.4 Numerical Implementation

Stationary configurations are obtained by solving $\nabla F_{\text{red}} = 0$ using multi-start root finding. Random initial conditions are sampled uniformly in a bounded region of (m, u, v) space, and solutions are accepted if the residual is below a fixed tolerance.

Due to the symmetry $v \mapsto -v$, solutions are reduced to a canonical representation using $|v|$, and numerically equivalent solutions are identified and removed.

For each (λ, η) , all distinct stationary points are collected, and the global minimum is selected by direct comparison of F_{red} . One-dimensional scans in η at fixed λ are performed to track the evolution of (m, u, v) and identify the phase boundary.

The numerical boundary scan confirms the analytic result $\eta_c(\lambda) = 0$ across $\lambda \in [0.5, 2.0]$.

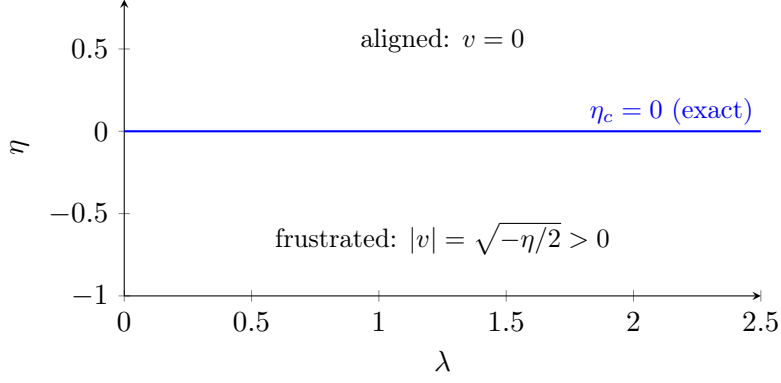


Figure 1: Phase diagram of the quartic toy model (5). The exact phase boundary $\eta_c(\lambda) = 0$ (blue) separates the aligned sector from the frustrated sector. The boundary is λ -independent, reflecting the decoupling of the longitudinal and transverse stationarity equations.

2.5 Phase Diagram

2.6 Remark on the Level-0 Reduction

The exact solvability of the quartic model is its main strength as a benchmark, but also its limitation: λ and η do not couple in this model, so the phase boundary carries no λ -dependent information. Genuine joint dependence $\eta_c(\lambda) \neq \text{const}$ requires a coupled interaction term, which appears naturally in the qubit state-structure model of Section 3.

3 Level 1: Qubit State-Structure Model

3.1 Structural Free Energy Functional

We promote the structural operator A to a dynamical variable and define

$$F_{\text{struct}}[\rho, A] = \text{Tr}(\rho H) - T_{\text{eff}} S[\rho] + \lambda \text{Tr}([A, \rho]^\dagger [A, \rho]) - \eta \text{Tr}(A\rho) - \frac{\mu_A}{2} \text{Tr}(A^2) + \frac{g_A}{4} \text{Tr}(A^4), \quad (11)$$

where $S[\rho] = -k_B \text{Tr}(\rho \ln \rho)$ is the von Neumann entropy. The commutator term $\text{Tr}([A, \rho]^\dagger [A, \rho])$ penalises structural incompatibility; the linear term $-\eta \text{Tr}(A\rho)$ energetically favours alignment.

3.2 Stationary Equations

Minimising F_{struct} under $\text{Tr}(\rho) = 1$ yields

$$H + k_B T_{\text{eff}} \ln \rho + 2\lambda [A, [A, \rho]] - \eta A + \tilde{\mu} I = 0, \quad (12)$$

$$2\lambda [\rho, [A, \rho]] - \eta \rho - \mu_A A + g_A A^3 = 0. \quad (13)$$

Eq. (12) generalises the Gibbs condition by a structural correction; Eq. (13) determines A self-consistently.

3.3 Qubit Realisation

With the parametrisation

$$H = \frac{\omega}{2} \sigma_z, \quad \rho = \frac{1}{2}(I + \vec{r} \cdot \vec{\sigma}), \quad A = \vec{a} \cdot \vec{\sigma}, \quad (14)$$

and the identities $[A, \rho] = i(\vec{a} \times \vec{r}) \cdot \vec{\sigma}$, $\text{Tr}([A, \rho]^\dagger [A, \rho]) = 2|\vec{a} \times \vec{r}|^2$, $\text{Tr}(A\rho) = \vec{a} \cdot \vec{r}$, the full free energy becomes

$$F_{\text{struct}}(\vec{r}, \vec{a}) = \frac{\omega}{2} r_z - T_{\text{eff}} S(r) + 2\lambda |\vec{a} \times \vec{r}|^2 - \eta \vec{a} \cdot \vec{r} - \frac{\mu_A}{2} |\vec{a}|^2 + \frac{g_A}{4} |\vec{a}|^4, \quad (15)$$

with $S(r) = -k_B[\frac{1+r}{2} \ln \frac{1+r}{2} + \frac{1-r}{2} \ln \frac{1-r}{2}]$.

3.4 Reduced Stationary System

Using the frustrated ansatz $\vec{r} = (0, 0, -m)$, $\vec{a} = (v, 0, -u)$, the reduced free energy is

$$F_{\text{red}}^{(1)}(m, u, v) = -\frac{\omega}{2}m - T_{\text{eff}}S(m) + 2\lambda v^2 m^2 - \eta u m - \frac{\mu_A}{2}(u^2 + v^2) + \frac{g_A}{4}(u^2 + v^2)^2, \quad (16)$$

and the stationarity equations read

$$-\frac{\omega}{2} + \frac{k_B T_{\text{eff}}}{2} \ln \frac{1+m}{1-m} + 4\lambda v^2 m - \eta u = 0, \quad (17)$$

$$-\eta m - \mu_A u + g_A(u^2 + v^2)u = 0, \quad (18)$$

$$v[4\lambda m^2 - \mu_A + g_A(u^2 + v^2)] = 0. \quad (19)$$

3.5 Comparison with the Level-0 Model

The Level-0 model (5) may be viewed as a phenomenological reduction of $F_{\text{red}}^{(1)}$, obtained by suppressing entropy and cross-coupling terms and by fixing effective Landau parameters.

The key difference is the $v^2 m^2$ coupling term in $F_{\text{red}}^{(1)}$: this term couples λ and v through the longitudinal order parameter m , and can generate a non-trivial λ -dependent phase boundary,

$$\eta_c(\lambda) \neq \text{const}, \quad (20)$$

in contrast to the exact $\eta_c(\lambda) = 0$ of the Level-0 reduction. Specifically, from Eq. (19), the frustrated branch ($v \neq 0$) requires

$$g_A(u^2 + v^2) = \mu_A - 4\lambda m^2, \quad (21)$$

so that increasing λ suppresses the transverse sector through the dynamical coupling to $m(\lambda, \eta)$.

3.6 Numerical Procedure for Level 1

For fixed parameters $(\omega, k_B T_{\text{eff}}, \mu_A, g_A)$ and couplings (λ, η) , the system (17)–(19) is solved by multi-start root finding.

1. Initialise at $(\lambda, \eta) = (0, 0)$ from the Gibbs value $m_0 = \tanh(\omega/(2k_B T_{\text{eff}}))$, $u_0 = v_0 = 0$.
2. Continue solutions across the parameter plane by nearest-neighbour initialisation.
3. Select the global minimum by evaluating $F_{\text{red}}^{(1)}$.

This procedure is expected to yield a λ -dependent phase boundary $\eta_c(\lambda)$ that reflects the nontrivial state–structure coupling absent in the Level-0 model. A full numerical exploration of the Level-1 model, including quantitative determination of $\eta_c(\lambda)$, is deferred to future work.

4 Numerical Pipeline and Results

4.1 Pipeline Overview

The numerical analysis is implemented in a five-step pipeline (`mftoe_phase_v2.py`), available in the companion repository:

<https://github.com/Chris4081/mftoe-proof>

1. **Branch tracking:** Multi-start root finding applied to $\nabla F_{\text{red}}(\lambda, \eta; m, u, v) = 0$ at each (λ, η) grid point. Solutions are deduplicated in the symmetry-reduced space $(m, u, |v|)$. Branches are retained if they span at least three η -values and are not energetically degenerate with an existing branch.
2. **Transition analysis:** For each λ -cut, global-minimum curves and pairwise free-energy crossings are identified. Branch switches are filtered: only switches satisfying $|\Delta F| < \varepsilon_F$ and $|\Delta m| > \varepsilon_m$ or $|\Delta v| > \varepsilon_v$ are recorded as genuine transitions.
3. **Phase diagram:** The (λ, η) plane is classified into four sectors (aligned/frustrated $\times$ single/multi minimum) using $\varepsilon_v = 10^{-3}$ as the frustration threshold.
4. **Boundary extraction and fit:** The critical line $\eta_c(\lambda)$ is extracted by linear interpolation between the last frustrated and first aligned point along each λ -cut. A refined local scan ($\lambda \in [0.5, 2.0]$, $\eta \in [-0.5, 0.1]$ at resolution $\Delta\eta = 0.005$) is used to confirm the expected relation $\eta_c(\lambda) = 0$.
5. **Overlay:** The numerically extracted boundary is superimposed on the phase diagram.

4.2 Connection to DESI DR2 Cosmology Tests

The quartic phase analysis presented in this paper is conceptually separate from the cosmological reconstruction pipeline, which provides an independent numerical test of MFToE-inspired vacuum dynamics against DESI DR2 BAO measurements [4, 5].

The main cosmological results are:

Model	Mode	χ^2	χ_{red}^2	w_{tot}
MFToE Baseline	targetH0	16.12	0.85	-0.282
MFToE Relaxion	relaxion	19.17	1.01	-0.289
Covariance $\rho = 0.3$	baseline	17.79	0.94	—
CAMB + $H_0 r_d$ match	joint	16.12	1.07	—

All tested models are compatible with DESI DR2 (2025) compressed BAO measurements at the level of reduced $\chi^2 \sim 1$ within the present compressed-data comparison. Deviations from Λ CDM remain below $\sim 1\%$ in both $H(z)$ and $d_L(z)$ within the explored parameter range.

4.3 Data Availability

The DESI DR2 BAO measurements used in the cosmological pipeline are publicly available under CC BY 4.0. The numerical code for the quartic phase analysis and the full cosmological pipeline is available in the companion repository [3]. All datasets are used in accordance with their respective licenses and cited according to collaboration guidelines.

5 Interpretation

The quartic model exhibits a simple separation of roles between the control parameters:

- λ controls the longitudinal ordering in m and u ,
- η controls the onset of the transverse component v .

Because the variables decouple at quadratic order, the location of the phase boundary is largely independent of λ . The exact result $\eta_c(\lambda) = 0$ is therefore a direct consequence of the separable structure of the model.

This highlights a key limitation of the present quartic reduction: nontrivial interplay between λ and η requires additional coupling terms, as present in the Level-1 qubit model of Section 3.

6 Summary

We analysed a minimal quartic state–structure model with three scalar order parameters. The full set of stationary points was obtained numerically, and the global minimum was used to construct a phase diagram.

The model exhibits two main regimes:

- a frustrated regime with $v \neq 0$ for $\eta < 0$,
- an aligned regime with $v \approx 0$ for $\eta \geq 0$.

The corresponding phase boundary satisfies $\eta_c(\lambda) = 0$, consistent with the separable analytic structure of the model.

A two-level summary of the models studied is as follows:

	Level 0 (quartic)	Level 1 (qubit)
Free energy	$-\lambda(m^2 + u^2) + \eta v^2 + \sum \phi^4$	full $F_{\text{struct}}[\rho, A]$
λ – η coupling	none (decoupled)	via $v^2 m^2$ coupling
Phase boundary	$\eta_c(\lambda) = 0$ (exact)	potentially λ -dependent (future study)
Numerical code	✓ implemented	formulated, future work

While the quartic model is intentionally minimal, it provides a transparent testbed for numerical phase selection and branch-tracking. Extensions with additional coupling terms — as present in the Level-1 qubit model — are required to generate nontrivial λ -dependent phase boundaries and constitute the next step in the programme.

References

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